Isotroic Elatic Model Soler

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1 Control Equantion

1.1 implicit mode

This problem can be solved by the implicit mode. While, a tangent matrix A_{ijkl} is needed in this solver. This is a boudary value problem, this question was controled by follow Eq. 1.

$$A_{ijkl}u_{k,l} = -X_{ij} + y_{ij} \tag{1}$$

1.2 explicit mode

If we want to solve the problem without the trangent matrix A_{ijkl} , then we need to use the Newtown's Second Law $F = m \cdot a$. The process is controlled by Eq.2 .

$$D_{ij}a_j = -X_{ij,j} + y_i \tag{2}$$

2 Elastic Matrix

This is the equation for the stiffness matrix Eq.2.

$$A_{ijkl} = \frac{\partial \sigma_{ij}}{\partial \epsilon_{kl}} \tag{3}$$

3 Solver

Algorithm 1 Framework of ensemble learning for our system.

Calculate the boundary condition, Dirichlet Boundary, Neuman Boundary
Applly boundary condition on the model
while $(rtol \leq err)$ and $(iter \leq iter_{max})$ **do**
    Calculate the trial displacement δu from the pde model
    Get strain from $\delta \epsilon_{ij} = \nabla \delta u$
    calculate stress $\delta \sigma_{ij} = A_{ijkl} \delta \epsilon_{kl}$
    renew the stress $\sigma_{ij} = \sigma_{ij} + \delta \sigma_{ij}$
    renew the domain $u_i = u_i + \delta u_i$
    calculate the err $err = \frac{\|\delta u\|}{\|u\|}$
end while

Input: The set of positive samples for current batch, P_n; The set of unlabelled samples for current batch, U_n; Ensemble of classifiers on former batches, E_{n-1};
Output: Ensemble of classifiers on the current batch, E_n;
Extracting the set of reliable negative and/or positive samples T_n from U_n with help of P_n;
Training ensemble of classifiers E on $T_n \cup P_n$, with help of data in former batches;
$E_n = E_{n-1} cup E$;
Classifying samples in $U_n - T_n$ by E_n;
Deleting some weak classifiers in E_n so as to keep the capacity of E_n;
return E_n;

for each $i \in [1, 9]$ **do**
    initialize a tree T_i with only a leaf (the root);
    $T = T \cup T_i$;
end for
for all c such that $c \in RecentMBatch(E_{n-1})$ **do**
    $T = T \cup PosSample(c)$;
end for;
for $i = 1$; $i < n$; $i++$ **do**
    // Your source here;
end for
for $i = 1$ to n **do**
    // Your source here;
end for
// Reusing recent base classifiers.
while $(|E_n| \leq L_1) and (D \neq \phi)$ **do**
    Selecting the most recent classifier c_i from D;
    $D = D - c_i$;
    $E_n = E_n + c_i$;
end while